ELEC026.23: Machine Learning
Final Exam | Project Specification

Title: Design and Development of a Company Website Showcasing Core Machine Learning Algorithms

Objective: The objective of this project is to develop a comprehensive company website that demonstrates the application of core machine learning algorithms, enabling students to understand and communicate the relevance of these algorithms in solving practical business or societal problems. The project encourages students to simulate a real-world business scenario, or societal problematic case, select and apply data and algorithms pertinent to their fictional company, and present their solutions through an informative and interactive website.

Project Description
Students are required to form groups of 4 members. Each group will conceptualize a fictional company in a chosen industry (e.g., retail, finance, healthcare, etc.) and design a company website that showcases the application of the following machine learning algorithms:

1. **Linear Regression**
2. **Naive Bayes**
3. **K-Nearest Neighbors (KNN)**
4. **Support Vector Machine**
5. **Decision Tree**
6. **Artificial Neural Network**

The website must include an overview of the company and its data-driven needs, detailed explanations of how each algorithm is relevant to their business operations, and sample datasets (either generated or openly available) used to illustrate the algorithms' applications.

Project Requirements
1. Company Concept and Website Content
    a. Create a fictional company with a unique name and industry focus.
    b. Develop a brief overview of the company's mission, vision, and operational goals.
    c. Explain how the company utilizes machine learning to solve specific industry-related challenges.
2. Algorithm Implementation and Explanation
    a. For each required algorithm:
        i. Describe its purpose and relevance to the company's business context.
        ii. Explain what data is used for the algorithm, why it is necessary, and how the algorithm contributes to the company’s goals.
        iii. Provide sample outputs or interpretations of the results.

    b. Ensure all algorithms are represented in ways that are both accessible and insightful for a non-technical audience, aligning explanations with real-world business scenarios.
3. Website Development
    a. Design a professional and user-friendly website that aligns with the company's branding.
    b. Include sections dedicated to each algorithm, with visualizations or interactive elements where possible.
    c. Make navigation intuitive, allowing users to easily find information on each algorithm and its application within the company.
    d. Consider incorporating multimedia (e.g., images, videos, charts) to enhance understanding.
4. Presentation and Q&A
    a. Prepare a presentation to demonstrate the website and explain the purpose of the algorithms and data used.

Grading Criteria

Total points: 50/50

Criterion	**Meeting** **10**	**Approaching** **8**	**Far from approaching** **6**
Website Aesthetics and Navigation (Aesthetic appeal, ease of navigation, and alignment with the company's branding)	The website is highly professional, visually appealing, and exhibits exceptional design coherence. Navigation is intuitive, seamless, and enhances the user's understanding of the company and its machine learning applications.	The website is functional and generally professional with clear design elements. Navigation is logical, allowing users to find information without significant difficulty.	The website lacks visual polish or is disorganized, detracting from the presentation of the content. Navigation is confusing, inconsistent, or contains significant usability issues.
Clarity and Accuracy of Algorithm Explanations (Clear and accurate explanations of each algorithm's purpose and application in the business context)	Explanations of the core machine learning algorithms are exceptionally clear, sophisticated, and completely accurate. The description effectively communicates complex concepts to	Explanations of the core machine learning algorithms are mostly clear and accurate, providing a sound understanding of their function. Minor inaccuracies or moments of technical jargon slightly detract	Explanations of the machine learning algorithms are vague, inaccurate, or contain significant technical errors. The content fails to clearly communicate how the algorithms work or their relevance to

	a general business or societal audience without oversimplification, demonstrating deep technical understanding.	from overall clarity.	the problem.
Relevance and Justification of Data and Algorithms (Justification of the chosen data for each algorithm, with insight into its business significance / problem being solved)	The selection of data and machine learning algorithms is perfectly aligned with the fictional company's objectives and the chosen problem, supported by a strong and rigorous justification. The rationale demonstrates a profound grasp of real-world data science constraints and opportunities.	The selected data and machine learning algorithms are appropriate for the company's problem, and a logical justification is provided. The connection between the problem, data, and algorithms is generally clear and sensible.	The selected data or machine learning algorithms are poorly matched to the company's problem, or the justification for their choice is weak, missing, or fundamentally flawed.
Depth of Algorithm Application and Results Interpretation (Appropriateness and clarity of algorithm selection, use, and results interpretation)	The website presents a detailed, insightful discussion of the algorithm's application, including a comprehensive and nuanced interpretation of potential results (e.g., performance metrics, business impact). The interpretation synthesizes technical findings with practical business or societal implications.	The website describes the application of the algorithm and provides a clear interpretation of the simulated or expected results. The interpretation adequately connects the technical outcomes to the practical context but may lack deeper analytical insight.	The website provides only a superficial description of the algorithm's application, or the interpretation of results is simplistic, missing, or fails to connect the technical outcomes back to the project's real-world problem.
Individual Presentation Quality and Depth of Understanding (Clarity and professionalism in presentation, ability	The individual presentation is delivered with outstanding clarity, professionalism, and confidence. Responses to questions are	The individual presentation is clear and well-structured, delivered with competence. Responses to questions are generally accurate	The individual presentation is poorly organized or lacks clarity, and/or the presenter appears unprepared. Responses to questions are vague,

to answer questions with depth and understanding)	insightful, demonstrating an expert-level, comprehensive grasp of the entire project, its technical aspects, and its business/societal relevance.	and demonstrate a solid working knowledge of their specific contributions and the overall project.	inaccurate, or reveal a significant lack of understanding of key project components or machine learning concepts.

IT3A/CS3

Group	**Company**	**Progress - October 23 5/5**
Tan, C Amoncio Narita Matol	Homewise AI - real-estate	3/5
Bancruz Liboon Navales Solis	Logistics	Absent: Navales 4/5
Reponoya Olpenado Palencia Rivera	RoastLab Coffee	4/5
Brua Tan, K Pavo Salibay	ShopSmart Analytics	Absent: Tan, K. 3/5
Eucare Allawan Mahipus Lopena	Learning simulations	3/5
Amper Ariego Dael Malaga	Military Analytics - triage and operations	4/5

IT3B

Group	**Company**	**Progress - October 22 5/5**
Fuentes Vergara Alconera Nanual	DineMind - business intelligence for restaurants	3/5
Militares Pelpinosas Espera	E-commerce	3/5
Ramos Galorio Lulab Corbita	RoomSense Technologies - building analysis	Absent: Lulab 2/5
Villacampa Gatan Mendoza Reyes	Agriculture	2/5
Millondaga Attos Eñola	Allonola - online shopping site	3/5
Ugay Navarro Abad Dimapilis	PesoLend - loan application	Absent: Abad 3/5